Dilmurod Valiev

AI/ML researcher — neuro-symbolic AI, 3D digital twins, speech interfaces

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PhD candidate building speech interfaces for 3D digital twins: a language model turns an instruction into a symbolic command, and a deterministic validator either executes it on the twin or refuses it with a reason. Production web and mobile work in React, Laravel and Flutter alongside the research. Looking for digital twin, industrial simulation or applied ML engineering roles.

EXPERIENCE

AI/ML researcher — Inha University AI Lab, Incheon Mar 2021 – present

- Built the speech-to-symbolic pipeline for aircraft-maintenance training on HoloLens: ASR, intent parsing and a symbolic command executed on a Unity digital twin.
- Designed the **action sandbox**, a deterministic validator that admits or refuses each command against twin state, procedure and user intent. Removing it raises unsafe actions from 0.7% to 11.8%.
- Extended speech recognition to Korean for the maintenance domain and benchmarked ASR and LLM back-ends for sequence-to-symbol generation.
- Co-authored the framework paper (*Sensors*, 2023); second paper under review at MDPI *Sensors* (2026).

Full-stack developer — Designmodo, Brooklyn, NY (remote) Sep 2021 – Jan 2022

- Cut render time **52%** on multi-component previews in a React editor.
- Built a SlateJS text editor from scratch, with independent undo/redo history for editing and for document actions.
- Migrated legacy HTML projects to JSON through Laravel queues, with WebSocket progress reporting; reduced image storage by 24% with versioned backups.

Flutter developer — Findme, Togliatti, Russia (remote) Aug 2021 – Jan 2022

- Shipped three consumer apps; improved rendering **27%** by narrowing widget rebuild scope.
- Added dark mode, custom animations and admin monitoring endpoints on a Laravel back end.

PUBLICATIONS

[1] Siyaev, A.; **Valiev, D.**; Jo, G.-S. Interaction with Industrial Digital Twin Using Neuro-Symbolic Reasoning. *Sensors* 2023, 23(3), 1729. doi:10.3390/s23031729

[2] **Valiev, D.**; Kim, Y.-S.; Jo, G.-S. Context-Aware Interaction with 3D Digital Twins Through Natural Language Speech Understanding. Under review, MDPI *Sensors*, 2026.

EDUCATION

Integrated MS/PhD, Inha University — Incheon. CGPA 4.20/4.5 Mar 2021 – present

BSc, Inha University in Tashkent. First of 270 students, 4.48/4.5. 2016 – 2020

Dean's list, graduated with honors, full scholarship.

SKILLS

Languages	Python, TypeScript / JavaScript, C#, SQL
ML	PyTorch, TensorFlow / Keras, Whisper, LLM fine-tuning, structured output, RAG evaluation
3D / XR	Unity, HoloLens / MRTK, Three.js, WebXR
Web	React, Next.js, Laravel, Flutter, FastAPI, Docker, GitHub Actions, PostgreSQL, MySQL, NoSQL